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Abstract

This report presents two data-science case studies that apply unsupervised and supervised learning to real datasets. The first segments the 3,999 members of the EastWest Airlines loyalty programme using K-Means, hierarchical clustering and DBSCAN. The data were standardised, the number of clusters was explored with the elbow method, linkage criteria and DBSCAN parameters (eps, minPts) were compared, and the results were evaluated with the silhouette score and PCA plots. K-Means with four clusters (silhouette 0.3092) produced balanced, interpretable segments: low-activity members, credit-card earners, frequent flyers and a small elite group. Average-linkage hierarchical clustering scored highest (0.7731) but placed 3,979 customers in a single cluster, and DBSCAN identified 98 unusual customers as noise, so K-Means is recommended for segmentation and DBSCAN for outlier detection.

The second study investigates which physicochemical properties explain the sensory quality of red (1,599) and white (4,898) vinho verde wines. Descriptive statistics, the coefficient of variation, correlation analysis, histograms, box plots and heatmaps showed that alcohol is the strongest correlate of quality ($r = 0.476$ red, 0.436 white), followed by volatile acidity (red) and density (white). Multiple linear regression models trained on an 80/20 split reached $R^2 = 0.4032$ and RMSE = 0.6245 for red wine and $R^2 = 0.2653$ and RMSE = 0.7543 for white wine, with multicollinearity between residual sugar, density and alcohol in white wine. The models can support preliminary laboratory screening, but they are not reliable enough to replace sensory evaluation. Together, the studies show how appropriate evaluation measures and careful interpretation turn data into practical decisions.

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CHAPTER 1
INTRODUCTION

1.1 Overview of the Case Studies

Data science turns raw records into decisions. This report applies two complementary techniques to two real datasets. Part A uses unsupervised learning (clustering) to discover natural customer groups in an airline loyalty programme, where no "correct" label exists. Part B uses supervised learning (multiple linear regression) to predict the sensory quality score of wine from laboratory measurements.

Table 1.1: Overview of the two case studies

	Part A – Customer Clustering	**Part B – Wine Quality**
Question	Which natural customer segments exist in an airline loyalty programme?	Which physicochemical properties explain wine quality, and can they predict it?
Dataset	EastWest Airlines loyalty data	UCI Wine Quality data (red and white vinho verde)
Records	3,999 customers, 11 numeric attributes (plus ID#)	1,599 red and 4,898 white wines, 11 inputs plus quality
Learning type	Unsupervised (no target variable)	Supervised regression (target: quality score)
Methods	K-Means, Hierarchical (Ward, complete, average), DBSCAN	Multiple linear regression, 80/20 split
Evaluation	Silhouette score, cluster profiles, PCA plots	MAE, MSE, RMSE, R^2, standardized coefficients
Focus	Parameter experiments and segment interpretation	Statistics, EDA, model reliability and business decision

1.2 Significance and Objectives

Part A shows how an airline can move from treating every member alike to recognising distinct groups, such as frequent flyers, credit-card earners and dormant members, so that offers can be targeted. Part B addresses a quality-control question: a company that relies on sensory tasting wants to know whether cheap laboratory measurements can explain or predict quality. The objectives are to:

- explore both datasets with descriptive statistics and visualisations and check data quality;
- scale the airline data and implement K-Means, hierarchical clustering and DBSCAN with several parameter settings (elbow curve, linkage criteria, epsilon and minPts);
- evaluate the clusters with the silhouette score, visualise them with PCA and interpret each segment;
- compute statistical measures (mean, median, mode, standard deviation, skewness, coefficient of variation and Pearson correlation) for the wine data;
- build and evaluate a multiple linear regression model for wine quality, identify influential predictors and discuss multicollinearity;
- translate both analyses into business recommendations and state their limitations.

1.3 Concepts used in Methodology

- **Exploratory data analysis:** summary statistics, box plots, histograms, correlation heatmaps and scatter plots.
- **Preprocessing:** data-quality checks, removal of the identifier, standardisation (z-scores) of features, and an 80/20 train–test split with a fixed random state.
- **Clustering algorithms:** K-Means (centroid-based, with the elbow method), agglomerative hierarchical clustering (Ward, complete and average linkage) and DBSCAN (density-based, with noise detection).
- **Evaluation:** silhouette score and cluster profiling for clustering; MAE, MSE, RMSE and R^2 for regression.
- **Interpretation:** PCA for two-dimensional visualisation, standardized regression coefficients, and checks for multicollinearity.

1.4 Outline of the Results

In Part A, K-Means with K = 4 split the 3,999 customers into segments of 2543, 1273, 168 and 15 members with a silhouette score of 0.3092. Average-linkage hierarchical clustering scored highest (0.7731) but isolated just 20 customers from one cluster of 3,979, and DBSCAN (eps = 2.5, minPts = 5) found three dense groups and flagged 98 unusual customers as noise. In Part B, alcohol is the attribute most strongly associated with quality in both wine types (r = 0.476 red, 0.436 white); the regression models reached R^2 = 0.4032 (red) and 0.2653 (white) on the test set, which is useful for screening but not enough to replace sensory evaluation. Chapters 2–5 present Part A, Chapters 6–9 present Part B and Chapter 10 concludes. All code and outputs are in the Jupyter notebooks clusteringanalysis.ipynb and winequality.ipynb.

CHAPTER 2
PART A – CLUSTERING: PROBLEM AND DATA EXPLORATION

2.1 Problem Statement

The objective of this assignment is to introduce the K-Means, hierarchical and DBSCAN clustering algorithms and to gain hands-on experience of applying them to a real-world dataset. The dataset belongs to EastWest Airlines, whose frequent-flyer programme records for each member the miles balance, miles earned through different credit cards and bonus transactions, flying activity in the last 12 months, time since enrolment and whether an award was claimed.

Problem: using these attributes alone, without any predefined labels, group the customers into homogeneous segments; experiment with the parameters of each algorithm (number of clusters K, linkage criterion, epsilon and minPts); evaluate the results with the silhouette score; and interpret the characteristics of every cluster.

2.2 Dataset and Summary Statistics

The sheet "data" of EastWestAirlines.xlsx contains 3,999 customers and 12 columns: a unique identifier (ID#) and the 11 numeric attributes described in Table 2.1.

Table 2.1: Airline dataset – attributes

Attribute	Description
Balance	Miles eligible for award travel
Qual_miles	Miles that qualify the member for top-flight status
cc1_miles, cc2_miles, cc3_miles	Miles earned with the three partner credit cards (coded bands 1–5)
Bonus_miles, Bonus_trans	Miles earned from non-flight bonus transactions and the number of such transactions
Flight_miles_12mo, Flight_trans_12	Flight miles and number of flight transactions in the last 12 months
Days_since_enroll	Days since the member enrolled in the programme
Award?	Binary flag: 1 if the member has claimed an award, 0 otherwise

Table 2.2: Airline dataset – summary statistics of the 11 features (n = 3,999)

Feature	Mean	Std	Min	Median	Max	Skewness
Balance	73,601	100,776	0	43,097	1,704,838	5.00
Qual_miles	144	774	0	0	11,148	7.51
cc1_miles	2.06	1.38	1.00	1.00	5.00	0.86
cc2_miles	1.01	0.15	1.00	1.00	3.00	11.21
cc3_miles	1.01	0.20	1.00	1.00	5.00	17.20
Bonus_miles	17,145	24,151	0	7,171	263,685	2.84
Bonus_trans	11.60	9.60	0.00	12.00	86.00	1.16
Flight_miles_12mo	460	1,400	0	0	30,817	7.45
Flight_trans_12	1.37	3.79	0.00	0.00	53.00	5.49
Days_since_enroll	4,119	2,065	2	4,096	8,296	0.12
Award?	0.37	0.48	0.00	0.00	1.00	0.54

Observations: almost every monetary or activity variable is strongly right-skewed. Balance has a mean of 73,601 miles but a maximum of 1,704,838, and Bonus_miles ranges up to 263,685. The median Flight_miles_12mo and Flight_trans_12 are both 0, which means that more than half of the members did not fly at all in the last 12 months, while a few members flew more than 30,000 miles. Only Days_since_enroll (skewness 0.12) is close to symmetric. 37% of members (mean of Award? = 0.37) have claimed an award.

2.3 Data Quality and Outliers

The dataset has no missing values and no duplicate customers; all columns are already numeric. ID# is only a label and was excluded from clustering, leaving 11 features. The code still converts all features to numeric and keeps a median-fill step as a safeguard, but it changed nothing.

Outliers were not removed. The extreme values of Balance, Bonus_miles and flight activity belong to real, high-value members, and deleting them would remove exactly the segments of greatest commercial interest. The algorithms were therefore run on the full data, and their sensitivity to these extreme points is discussed in Chapters 4 and 5.

2.4 Visual Analysis

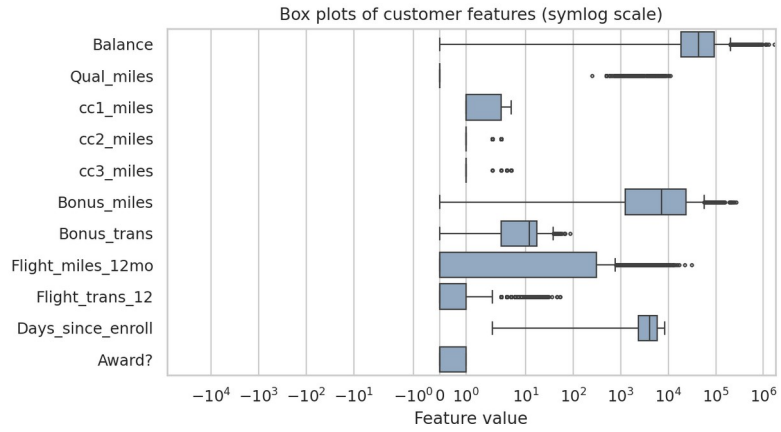


Figure 2.1: Airline data – box plots of customer features (symmetric-log scale)

Interpretation: the box plots confirm the skewness in Table 2.2. Balance, Bonus_miles, Flight_miles_12mo, Qual_miles and Flight_trans_12 have long chains of points above the upper whisker, so the boxes are compressed near zero. The credit-card columns take only a few coded values. These extreme points are the reason why feature scaling is required.

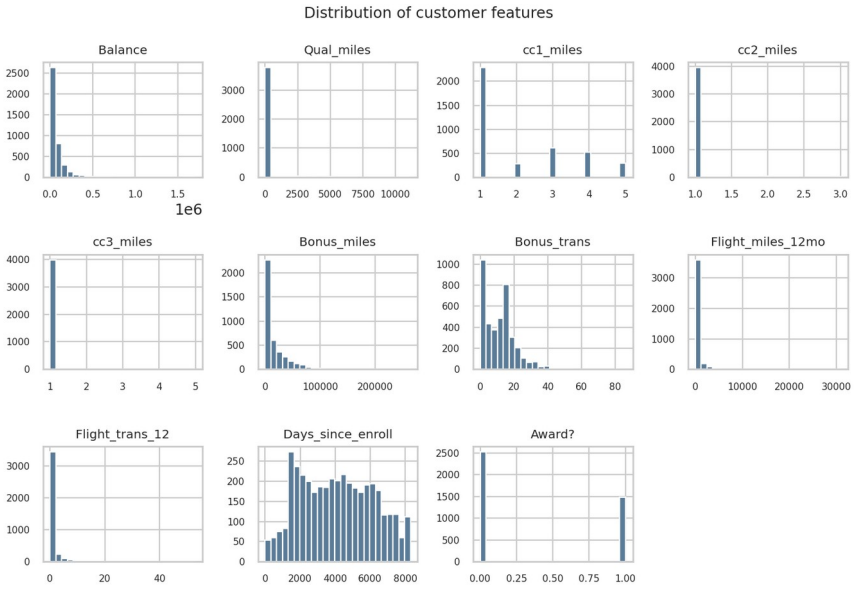


Figure 2.2: Airline data – distribution of customer features

Interpretation: Balance, Bonus_miles, Bonus_trans, Qual_miles and the flight variables pile up at small values with a long right tail; cc2_miles and cc3_miles are almost constant (value 1 for most members) and Award? is binary. Days_since_enroll is spread broadly across the range, so membership length is fairly evenly distributed.

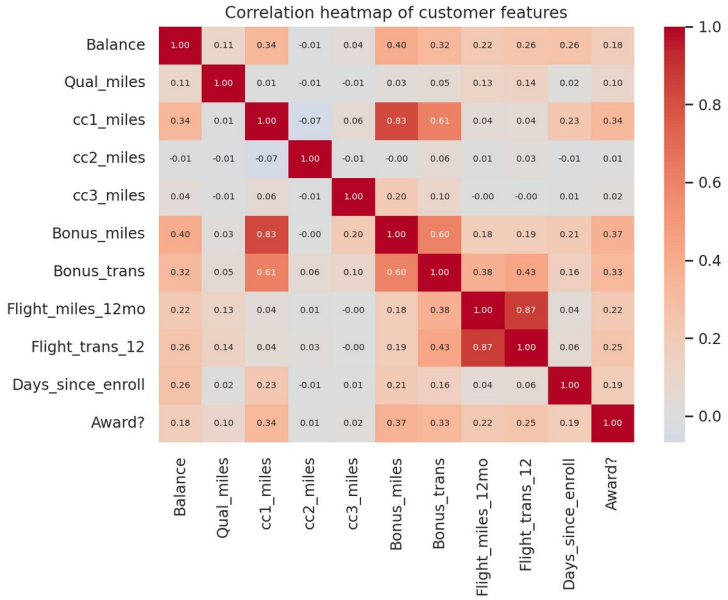


Figure 2.3: Airline data – correlation heatmap of customer features

Interpretation: the strongest relationships are Flight_miles_12mo and Flight_trans_12 (r = 0.87), cc1_miles and Bonus_miles (0.83), cc1_miles and Bonus_trans (0.61) and Bonus_miles and Bonus_trans (0.60). These pairs describe the same behaviour (flying, and earning miles through the first credit card), which suggests that two or three underlying dimensions, “flying activity” and “credit-card/bonus earning”, drive most of the structure that clustering will find.

Key findings: the data are heavy-tailed, flying activity and card-based earning are two distinct behaviours, and a large share of members are inactive. All three observations affect how the clustering results must be interpreted.

CHAPTER 3
PART A – CLUSTERING: PREPROCESSING AND K-MEANS

3.1 Project Workflow

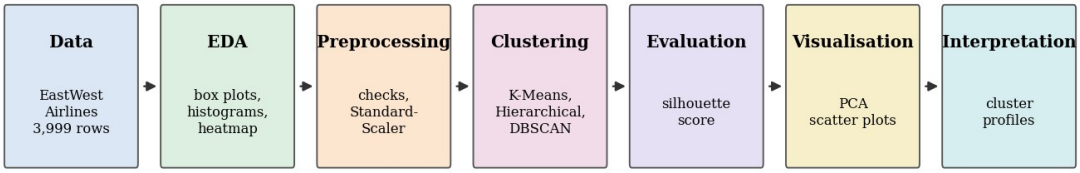


Figure 3.1: Workflow followed in the clustering case study

3.2 Data Preprocessing

Table 3.1: Airline data – preprocessing steps

Step	Columns	Method	Reason
Remove identifier	ID#	Dropped	A label, not a customer characteristic
Type check / missing values	All 11 features	Numeric conversion; median fill (none needed)	No missing values were found
Scaling	All 11 features	StandardScaler (z-score)	Balance reaches 1.7 million while cc2_miles ranges 1–3; distance-based methods would otherwise be dominated by large-valued features
Outliers	Balance, Bonus_miles, flight variables	Retained	They are genuine high-value customers (Section 2.3)

After scaling, each feature has mean 0 and standard deviation 1, giving a 3,999 × 11 matrix that is used by all three algorithms.

3.3 K-Means Clustering and the Elbow Method

K-Means assigns each customer to the nearest of K centroids and then recomputes the centroids until the assignments stop changing, minimising the within-cluster sum of squared distances (inertia). K must be chosen in advance, so K = 2 to 10 was tried (random_state = 42, n_init = 10) and the inertia and silhouette score were recorded.

Table 3.2: K-Means – inertia and silhouette score for different K

K	Inertia	Silhouette score
2	35,409.5	0.3254
3	30,906.1	0.3111
4	27,154.3	0.3092
5	24,541.5	0.3189
6	21,089.7	0.3334
7	18,891.9	0.2837
8	17,316.8	0.2367
9	15,961.5	0.2460
10	14,728.3	0.2621

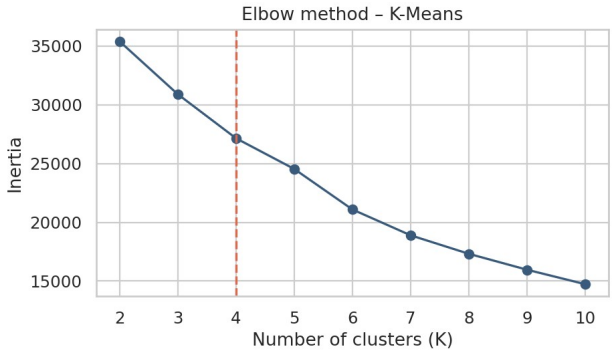


Figure 3.2: K-Means – elbow curve (dashed line at K = 4)

Interpretation: inertia falls steadily from 35,410 at K = 2 to 27,154 at K = 4 and then declines more slowly, so there is no sharp elbow, only a gradual bend around K = 4 to 5. The silhouette score stays within 0.31–0.33 for K = 2 to 6, so it does not strongly favour any value. K = 4 was selected as a compromise between compactness and interpretability: it is small enough to describe each segment, and it gives a silhouette score of 0.3092, a modest but real cluster structure.

3.4 K-Means Cluster Profiles

Table 3.3: K-Means (K = 4) – cluster sizes and mean feature values

Feature	Cluster 0	Cluster 1	Cluster 2	Cluster 3	All
Customers	15 (0.4%)	2,543 (63.6%)	1,273 (31.8%)	168 (4.2%)	3,999
Balance	138,061	43,794	116,817	191,574	73,601
Qual_miles	79	97	155	776	144
cc1_miles	3.47	1.22	3.70	2.17	2.06
cc2_miles	1.00	1.02	1.00	1.04	1.01
cc3_miles	4.07	1.00	1.00	1.00	1.01
Bonus_miles	93,928	4,701	39,216	31,412	17,145
Bonus_trans	28.07	6.92	18.76	26.78	11.60
Flight_miles_12mo	507	199	334	5,366	460
Flight_trans_12	1.60	0.62	1.00	15.64	1.37
Days_since_enroll	4,614	3,692	4,893	4,666	4,119
Award?	0.53	0.20	0.65	0.81	0.37

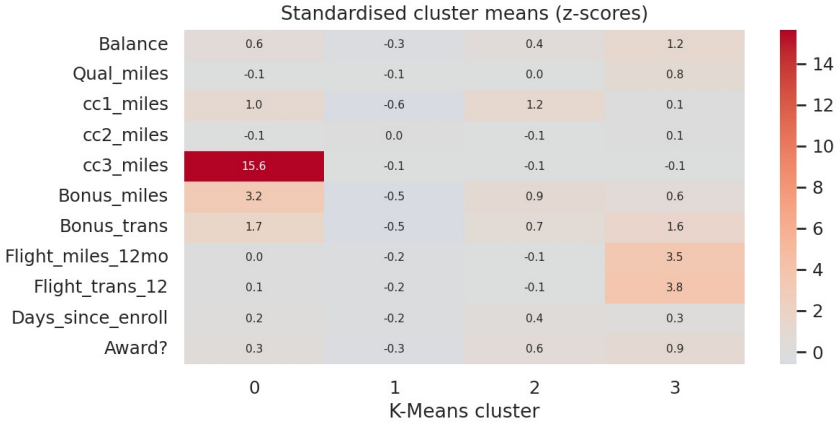


Figure 3.3: K-Means – standardised cluster means (z-scores)

Interpretation of the four segments:

- **Cluster 1 – Low-activity members (2543, 63.6%):** the largest group has the lowest balance (43,794), almost no bonus miles (4,701), less than one flight transaction a year and the lowest award rate (20%). They are mostly dormant or entry-level members.
- **Cluster 2 – Credit-card earners (1273, 31.8%):** high balance (116,817) and bonus miles (39,216) with the highest cc1_miles (3.70), but only one flight transaction a year. Their miles come from card spending rather than flying; 65% have claimed an award.
- **Cluster 3 – Frequent flyers (168, 4.2%):** by far the most flying activity (5,366 flight miles and 15.6 flight transactions a year), the highest qualifying miles (776) and an 81% award rate. They are the most engaged and valuable travellers.
- **Cluster 0 – Elite bonus-mile accumulators (15, 0.4%):** a very small group with very high bonus miles (93,928), the highest cc3_miles (4.07, about 15 standard deviations above the overall mean, so this card is what chiefly defines the cluster) and a high balance (138,061); 53% have claimed an award. They represent heavy users of partner cards.

CHAPTER 4
PART A – CLUSTERING: HIERARCHICAL CLUSTERING AND DBSCAN

4.1 Hierarchical Clustering

Agglomerative hierarchical clustering starts with every customer in its own cluster and repeatedly merges the two closest clusters. The meaning of "closest" is the linkage criterion: Ward linkage merges the pair that least increases the within-cluster variance, complete linkage uses the largest distance between members, and average linkage uses the mean distance. All three criteria were tested on the scaled data and cut at four clusters so that they are comparable with K-Means.

Table 4.1: Hierarchical clustering – linkage comparison (4 clusters)

Linkage	Cluster sizes (largest to smallest)	Silhouette score
Ward	2,446 / 1,380 / 130 / 43	0.3081
Complete	3,952 / 28 / 15 / 4	0.5986
Average	3,979 / 15 / 4 / 1	0.7731

Interpretation: Ward linkage produces balanced clusters and a silhouette score (0.3081) almost identical to K-Means (0.3092). Complete and average linkage obtain much higher silhouette scores, but only because they put almost every customer (3,952 and 3,979) into one giant cluster and isolate a handful of extreme customers in the others, including clusters of a single customer. The high score reflects the well-separated outliers rather than a useful segmentation, so the number alone is misleading.

4.2 DBSCAN Clustering

DBSCAN groups points that lie in dense regions. A customer is a core point if at least minPts customers lie within distance eps; clusters grow from core points, and points that belong to no dense region are labelled as noise (−1). Unlike K-Means, it needs no value of K and can detect outliers, but it is sensitive to eps and minPts. The k-distance plot (distance to the 5th nearest neighbour) in Figure 4.1 was used to choose a range of eps values; the sorted curve rises steeply in its upper tail, which marks the sparse, outlying customers.

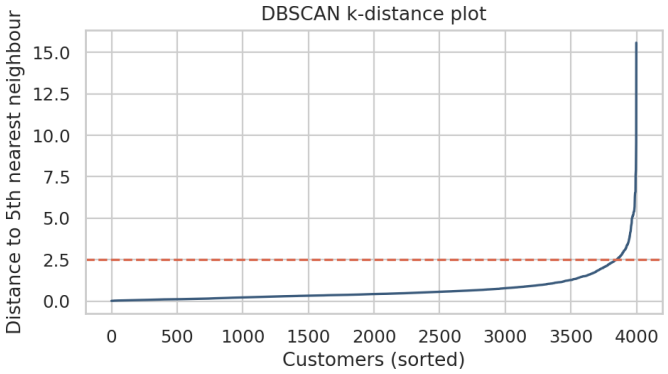


Figure 4.1: DBSCAN – sorted distance to the 5th nearest neighbour (dashed line at eps = 2.5)

Table 4.2: DBSCAN – parameter experiments on the scaled data

eps	minPts	Clusters	Noise points	Silhouette (excluding noise)
1.0	5	6	554	0.2817
1.0	10	3	675	0.3179
1.5	5	5	310	0.2940
1.5	10	3	377	0.2982
2.0	5	6	179	0.2805
2.0	10	3	237	0.2868
2.5	5	3	98	0.5604
2.5	10	4	136	0.5379
3.0	5	3	67	0.5542
3.0	10	3	81	0.5589

Interpretation: small eps values (1.0–2.0) label 179 to 675 customers as noise and split the dense core into several clusters with silhouette scores of only 0.28–0.32. From eps = 2.5 the dense core merges into one cluster, the noise falls to 98 or fewer customers and the silhouette score rises to 0.54–0.56. The setting eps = 2.5, minPts = 5 has the highest silhouette in the grid (0.5604) and was selected. It yields three clusters of 3,865, 23 and 13 customers and 98 noise points (2.5% of customers). The two small clusters are distinguished by the cc2_miles code (2 and 3 respectively) and have no qualifying miles, while the noise points are the high-value members, with a mean balance of 264,733, 4,762 flight miles a year and a 78% award rate.

CHAPTER 5
PART A – CLUSTERING: EVALUATION, VISUALISATION AND INTERPRETATION

5.1 Silhouette Comparison of the Algorithms

The silhouette score of a customer compares its average distance to its own cluster with that to the nearest other cluster. It ranges from −1 to +1; higher values mean tighter, better-separated clusters. Table 5.1 and Figure 5.1 compare all configurations.

Table 5.1: Airline data – comparison of clustering configurations

Algorithm	Clusters	Noise points	Silhouette score
K-Means (K=4)	4	0	0.3092
Hierarchical (Ward)	4	0	0.3081
Hierarchical (Complete)	4	0	0.5986
Hierarchical (Average)	4	0	0.7731
DBSCAN (eps=2.5, minPts=5)	3	98	0.5604

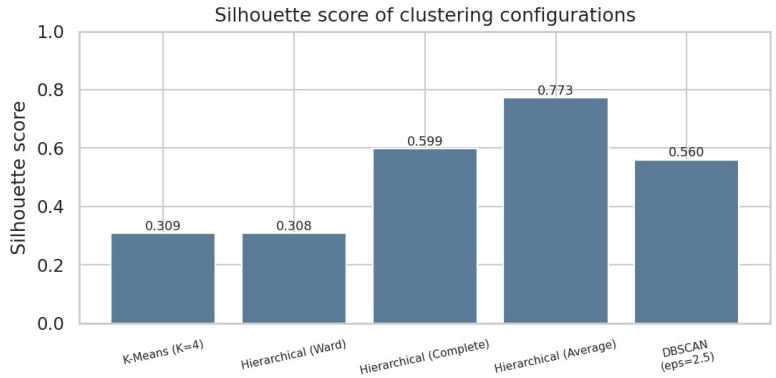


Figure 5.1: Silhouette score of the clustering configurations

Interpretation: the ranking by score is Hierarchical (average) 0.7731, Hierarchical (complete) 0.5986, DBSCAN 0.5604, K-Means 0.3092 and Hierarchical (Ward) 0.3081. This ranking must be read with care. Average and complete linkage contain one cluster with 99% of the customers; the DBSCAN score is computed after removing the 98 noise points and is dominated by its 3,865-customer cluster. Only K-Means and Ward score all customers with balanced groups, and both reach about 0.31, which indicates moderate, overlapping structure, as expected for behavioural data without clear gaps between customer types.

5.2 PCA Visualisation

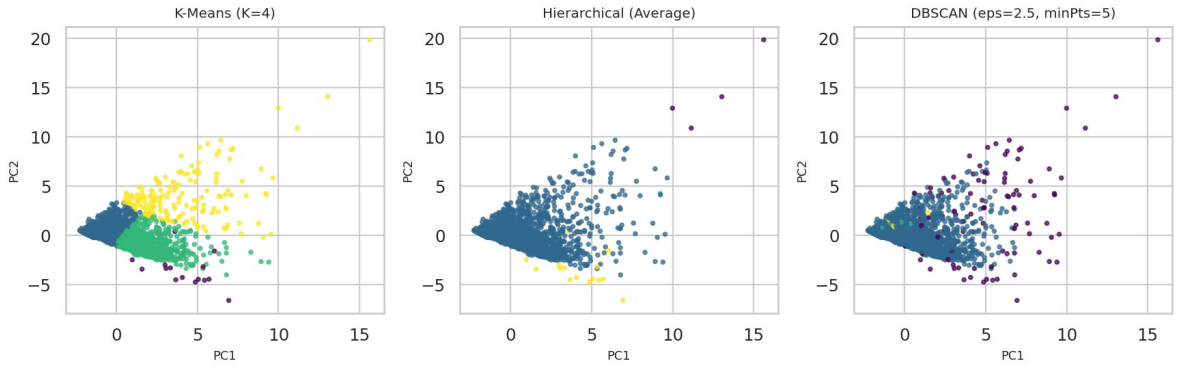


Figure 5.2: PCA projection of the clustering results (K-Means, hierarchical average, DBSCAN)

Interpretation: the first two principal components explain 29.9% and 15.7% of the variance (45.6% in total), so the two-dimensional view is only a partial picture. In the K-Means panel the four colours form a clear gradient from the dense mass of customers to the extended arms of high-activity customers. In the hierarchical panel almost all points share one colour, with a few isolated outliers, and in the DBSCAN panel the single dominant cluster is surrounded by the noise points, which lie on the periphery.

5.3 Characteristics of the Clusters

Table 5.2: Hierarchical (average) and DBSCAN – cluster sizes and key means

Algorithm / cluster	Customers	Balance	Bonus_miles	Flight_miles_12 mo	Award?
Hierarchical – cluster 0	4	132,000	65,634	19,960	1.00
Hierarchical – cluster 1	3,979	72,890	16,807	439	0.37
Hierarchical – cluster 2	1	1,704,838	17,108	4,823	1.00
Hierarchical – cluster 3	15	138,061	93,928	507	0.53
DBSCAN – noise	98	264,733	62,033	4,762	0.78
DBSCAN – cluster 0	3,865	69,065	16,058	353	0.36
DBSCAN – cluster 1	23	39,644	8,962	222	0.30
DBSCAN – cluster 2	13	41,562	16,338	323	0.31

Insights:

- **K-Means** gives four balanced, interpretable segments (low-activity, credit-card earners, frequent flyers and a small elite group) that map directly to marketing actions.
- **Hierarchical (average)** behaves as an outlier detector. Its four-member cluster has the most active customers (49 flight transactions and 69 bonus transactions on average, all with awards), and its single-customer cluster has a balance of 1.7 million miles. The remaining 3,979 customers are not separated.
- **DBSCAN** treats the bulk of members as one dense group and identifies 98 exceptional customers as noise (very high balances, qualifying miles and flying). This is useful for finding VIPs and data anomalies, but it does not segment the main customer base.

5.4 Best-Performing Approach and Recommendations

For customer segmentation, **K-Means with K = 4** is the most suitable model. Its clusters are balanced and well described by business-relevant behaviour, and its silhouette score (0.3092) is honestly computed on all customers. Average-linkage clustering has the highest silhouette score but fails the practical requirement of producing usable groups. DBSCAN is best used as a complement, to flag the 98 outlying customers for special treatment. The recommendations are:

- **Low-activity members (64% of customers):** run low-cost re-engagement campaigns, such as bonus-mile offers for a first flight or card activation, to move them into the active segments.
- **Credit-card earners (32%):** offer promotions that convert accumulated miles into flights, because their balances are high but their flying is low.
- **Frequent flyers (4%) and the elite group:** protect them with retention benefits, priority services and status-based rewards, since they generate most of the flight revenue.
- **DBSCAN noise customers:** review them individually for VIP programmes or for data-quality problems.

5.5 Limitations

- **Silhouette score is not enough:** very different cluster shapes can give similar scores, and the highest scores here belong to degenerate solutions. Business usefulness and cluster sizes must also be examined.
- **Skewed, outlier-heavy data and coded variables:** outliers were retained, which pulls centroids and creates tiny clusters. The cc1–cc3 bands and the binary Award? flag were treated as continuous after scaling, which is a simplification.
- **Parameter sensitivity and partial visualisation:** K, eps and minPts were chosen from limited grids, and PCA shows only about 46% of the variance, so the plots do not fully reflect the clustering in 11 dimensions.

CHAPTER 6
PART B – WINE QUALITY: PROBLEM, DATA AND STATISTICAL ANALYSIS

6.1 Case Study and Problem Statement

A wine company currently relies on sensory quality assessments carried out by tasters. Management wants to investigate whether physicochemical laboratory measurements can help explain differences in wine quality and support a data-driven quality assessment process.

Problem: analyse the wine quality dataset to determine which physicochemical characteristics are most strongly associated with quality, develop and evaluate a regression model that predicts the quality score from the laboratory measurements, and advise on a laboratory-based quality screening process.

6.2 Dataset Description

The UCI Wine Quality dataset contains two files from the Portuguese “vinho verde” wines: winequality-red.csv (1,599 wines) and winequality-white.csv (4,898 wines), both semicolon-delimited. Each wine has 11 laboratory measurements and a sensory quality score (an integer, in practice between 3 and 9). The two wine types differ in chemistry, so they were analysed separately.

Table 6.1: Wine dataset – attributes

Attribute	Role	Type
fixed acidity, volatile acidity, citric acid	Predictors (acidity)	Numeric
residual sugar, chlorides, sulphates	Predictors (composition)	Numeric
free sulfur dioxide, total sulfur dioxide	Predictors (preservatives)	Numeric
density, pH, alcohol	Predictors (physical properties)	Numeric
quality	Target – sensory score	Integer

6.3 Data Quality

Table 6.2: Wine dataset – size and data quality

Dataset	Rows × columns	Missing values	Duplicate rows
Red wine	1,599 × 12	0	240
White wine	4,898 × 12	0	937

Neither file has missing values. Duplicate rows exist (240 red, 937 white); they may be repeated measurements of similar wines, and they were retained in the analysis as in the notebook. Their possible effect on the evaluation is discussed in Section 9.3.

6.4 Task 1 – Statistical Analysis

Table 6.3 shows the mean, median, mode, standard deviation and skewness of alcohol, volatile acidity and residual sugar for each wine type.

Table 6.3: Wine – descriptive statistics

Wine	Attribute	Mean	Median	Mode	Std dev	Skewness
Red	alcohol	10.423	10.200	9.500	1.066	0.861
Red	volatile acidity	0.528	0.520	0.600	0.179	0.672
Red	residual sugar	2.539	2.200	2.000	1.410	4.541
White	alcohol	10.514	10.400	9.400	1.231	0.487
White	volatile acidity	0.278	0.260	0.280	0.101	1.577
White	residual sugar	6.391	5.200	1.200	5.072	1.077

Interpretation: all three attributes have mean greater than median and positive skewness, so their distributions have longer right tails. Alcohol is only mildly skewed (0.49–0.86). Red wine has a higher average volatile acidity (0.528 against 0.278), while white wine has far more residual sugar (6.39 against 2.54). Residual sugar in red wine is extremely skewed (4.54): most red wines are dry, but a few contain much more sugar.

Relative variability. The standard deviations are measured in different units, so the coefficient of variation (CV = standard deviation ÷ mean × 100) is used for comparison.

Table 6.4: Wine – coefficient of variation and Pearson correlation of alcohol with quality

Wine	CV alcohol	CV volatile acidity	CV residual sugar	r (alcohol, quality)
Red	10.22%	33.92%	55.54%	0.476
White	11.70%	36.23%	79.36%	0.436

Interpretation: residual sugar has the greatest relative variability in both wine types (55.5% for red and 79.4% for white), followed by volatile acidity, while alcohol is the most stable attribute (about 10–12%). Pearson's coefficient between alcohol and quality is +0.476 for red and +0.436 for white wine. The direction is positive (wines with more alcohol tend to receive higher scores) and the strength is moderate, since r lies between 0.4 and 0.5. Alcohol explains only about 19–23% of the variance in quality on its own ($r^2 = 0.23$ and 0.19), so other attributes also matter.

CHAPTER 7
PART B – WINE QUALITY: EXPLORATORY DATA ANALYSIS

7.1 Distribution of Quality Scores

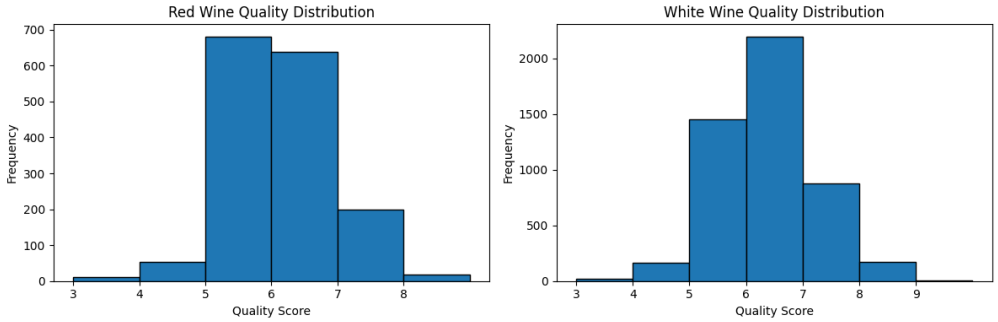


Figure 7.1: Wine – histogram of quality scores (red and white)

Interpretation: both distributions are roughly bell-shaped and concentrated in the middle of the scale. Most red wines score 5 (about 680 wines) or 6 (about 640), which is over 80% of the file; for white wine, scores 5, 6 and 7 cover more than 90%, with 6 the most common. Scores of 3, 4, 8 and 9 are rare (white wine has only 5 wines with 9, and the red file has no 9). The scale is therefore narrow in practice and heavily imbalanced, which makes extreme-quality wines hard to predict.

7.2 Alcohol Content by Quality Score

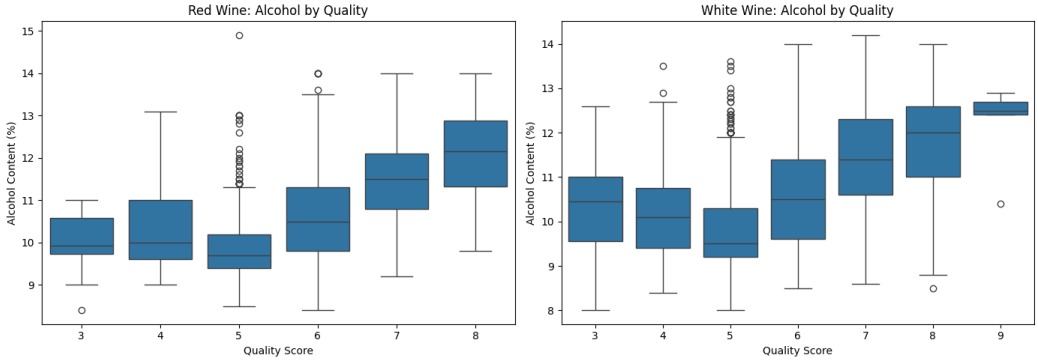


Figure 7.2: Wine – box plots of alcohol content grouped by quality score

Interpretation: from quality 5 upward the median alcohol rises steadily: in red wine from about 9.7% at score 5 to about 11.5% at score 7 and 12.2% at score 8, and in white wine from about 9.5% at score 5 to 11.4% at score 7, 12% at score 8 and 12.5% at score 9. The lowest scores (3 and 4) have slightly higher medians than score 5, so the pattern is not perfectly monotonic. Many high-alcohol outliers appear at score 5 (red up to about 15%, and a long row for white wine) and at red score 6, showing that some high-alcohol wines are still rated as average. The boxes of neighbouring scores overlap, so alcohol alone cannot separate the quality levels.

7.3 Correlation Heatmaps

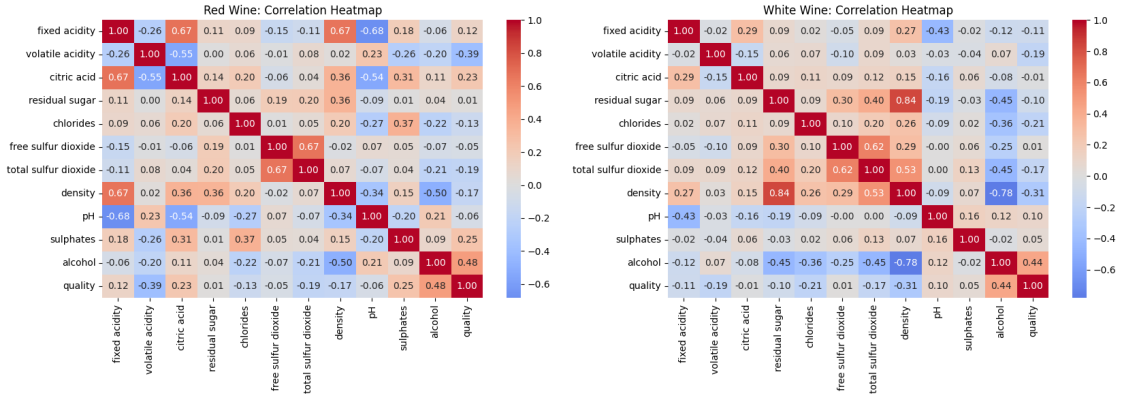


Figure 7.3: Wine – correlation heatmaps for red and white wine

Interpretation: in red wine, alcohol (+0.48), sulphates (+0.25) and citric acid (+0.23) are positively related to quality, and volatile acidity (−0.39) and total sulfur dioxide (−0.19) negatively. In white wine, alcohol (+0.44) is the main positive factor, while density (−0.31), chlorides (−0.21) and volatile acidity (−0.19) are negative. Among the predictors, white wine shows very strong links between residual sugar and density (+0.84) and between density and alcohol (−0.78); red wine has moderate links such as fixed acidity with pH (−0.68), citric acid (+0.67) and density (+0.67).

7.4 Attributes Most Strongly Associated with Quality

Table 7.1: Wine – two attributes with the largest absolute correlation with quality

Wine	Rank	Attribute	Pearson r	Direction
Red	1	alcohol	+0.476	Positive
Red	2	volatile acidity	−0.391	Negative
White	1	alcohol	+0.436	Positive
White	2	density	−0.307	Negative

Key findings: alcohol is the strongest single correlate of quality for both wine types, with a positive association. The second attribute differs: volatile acidity for red wine (higher values accompany lower scores, consistent with its link to unpleasant, vinegar-like flavours) and density for white wine (denser wines score lower, partly because density falls as alcohol rises). Correlation shows association only, not causation.

CHAPTER 8
PART B – WINE QUALITY: PREDICTIVE MODELLING

8.1 Model Development

A multiple linear regression model, quality = $\beta_0 + \beta_1 x_1 + \ldots + \beta_{11} x_{11}$, was fitted separately for red and white wine using all 11 physicochemical attributes. Each dataset was split 80% for training and 20% for testing with random_state = 42: 1,279 training and 320 test wines for red, and 3,918 training and 980 test wines for white. The model was fitted on the training set only and all metrics below are calculated on the test set.

8.2 Test-Set Performance

Table 8.1: Wine – regression performance on the test set

Metric	Red wine	White wine	Meaning
MAE	0.5035	0.5863	Average absolute error in quality points
MSE	0.3900	0.5690	Average squared error
RMSE	0.6245	0.7543	Typical error, in quality points
R^2	0.4032	0.2653	Share of variance in quality explained

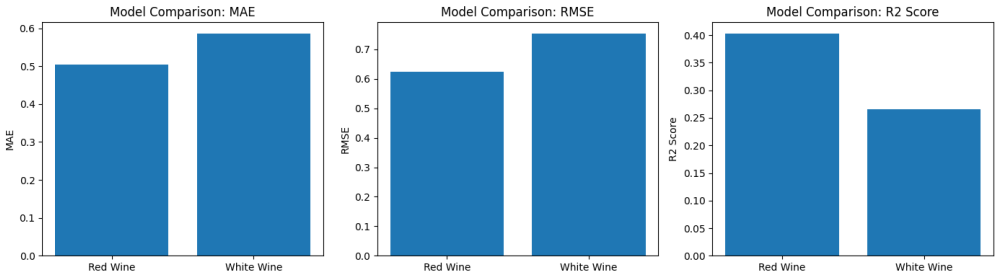


Figure 8.1: Wine – comparison of MAE, RMSE and R^2 for red and white models

Interpretation: the red wine model is better on every metric. On average its predictions are 0.50 quality points from the true score (RMSE 0.62) and it explains 40.3% of the variation. The white wine model has an average error of 0.59 points (RMSE 0.75) and explains 26.5%. More than half of the variation in quality is therefore not explained by the laboratory measurements in either model.

8.3 Influential Predictors and Multicollinearity

Raw coefficients cannot be compared because the attributes have different units. Both the predictors and the target were standardised, and the standardized coefficients show the change in quality (in standard deviations) per one standard deviation change in an attribute, holding the others constant.

Table 8.2: Wine – standardized regression coefficients

Attribute	Red wine	White wine
alcohol	0.3678	0.3171
volatile acidity	−0.2225	−0.2194
sulphates	0.1820	0.0831
total sulfur dioxide	−0.1488	−0.0115
chlorides	−0.1104	−0.0007
pH	−0.0751	0.1017
free sulfur dioxide	0.0719	0.0963
fixed acidity	0.0493	0.0437
citric acid	−0.0341	−0.0083
density	−0.0238	−0.4234
residual sugar	0.0117	0.4128

Multicollinearity check (before interpreting the effects). Strongly correlated predictors make coefficients unstable, because the model cannot separate their individual effects. Predictor pairs with an absolute correlation of at least 0.70 were therefore searched for.

Table 8.3: Wine – highly correlated predictor pairs ($|r| \geq 0.70$)

Wine	Attribute 1	Attribute 2	Absolute correlation
Red	–	–	No pair reaches 0.70
White	residual sugar	density	0.839
White	density	alcohol	0.780

Interpretation: for red wine no predictor pair reaches 0.70 (the largest moderate correlations are about 0.67–0.68), so the coefficients are reasonably stable. The two most influential predictors are **alcohol (+0.368)** and **volatile acidity (−0.222)**: higher alcohol raises, and higher volatile acidity lowers, the predicted quality. They agree with the correlation analysis in Section 7.4.

For white wine the two most influential predictors by absolute value are **density (−0.423)** and **residual sugar (+0.413)**, with alcohol third (+0.317). These are exactly the attributes involved in the strong collinearity (sugar–density 0.84, density–alcohol 0.78). Because sugar raises density and alcohol lowers it, the model assigns a large negative weight to density and a large positive weight to sugar that largely offset each other. These two coefficients should be read as a pair, not as independent effects, and a positive effect of sugar on quality contradicts its near-zero simple correlation (−0.10). Alcohol, whose simple correlation with quality (+0.44) is consistent with its coefficient, is the more dependable message for white wine. A variance inflation factor (VIF) analysis was not run and is suggested as future work.

8.4 Actual versus Predicted Quality

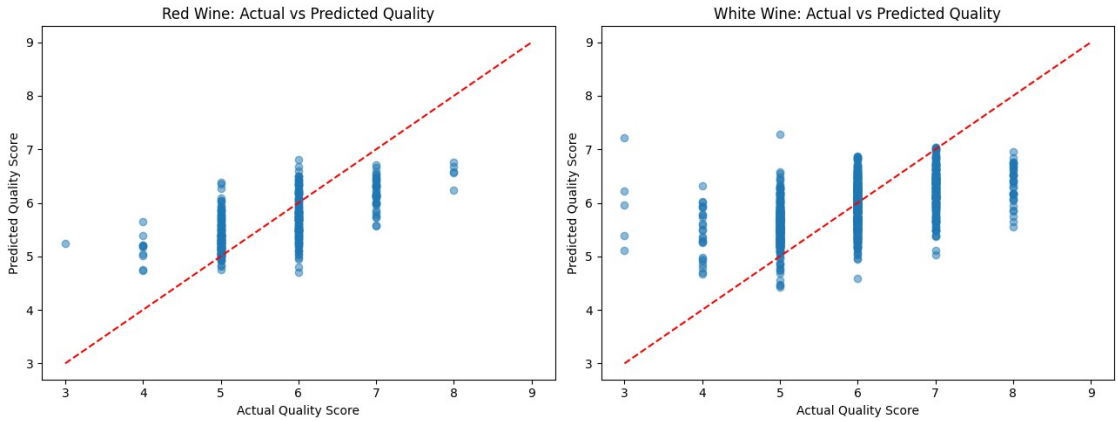


Figure 8.2: Wine – actual versus predicted quality on the test set (dashed line = perfect prediction)

Interpretation and limitations of the model: if the model were perfect, all points would lie on the dashed diagonal. Instead, the predictions for both wine types are confined to a narrow band (roughly 4.7–7 for red and 4.5–7.3 for white), whatever the true score. Wines with true scores of 3, 4 and 8 are pulled towards the middle: for example, red wines scored 8 are predicted around 6.2–6.8, and wines scored 3 are predicted near 5. The model captures the overall upward trend, since the predictions for score 7 are higher than for score 5, but each actual score is associated with a wide vertical spread of predictions, especially for white wine. The reasons are the limited R^2, the strong imbalance of the quality scores (few extreme wines to learn from), the assumption of a linear relationship, and the fact that sensory scores contain human subjectivity that laboratory data cannot capture. Predicting an integer score with a continuous output also means that errors of about half a point are unavoidable.

CHAPTER 9
PART B – WINE QUALITY: BUSINESS DECISION-MAKING

9.1 Measurements Deserving Attention in Screening

The company is considering a laboratory-based quality screening process. On the evidence of the correlation analysis and the standardized coefficients, two measurements deserve attention:

- **Alcohol content (both wine types):** it is the strongest correlate of quality (r = +0.476 red, +0.436 white), the largest standardized coefficient for red wine (+0.368) and an important predictor for white wine (+0.317). Lots with unusually low alcohol should be flagged for closer tasting.
- **Volatile acidity for red wine and density for white wine:** volatile acidity is the second most influential predictor of red wine quality (−0.222; r = −0.391), so high values should trigger review. For white wine, density has the second-largest correlation (−0.307) and the largest coefficient, but it is collinear with sugar and alcohol, so it should be monitored together with them rather than used alone.

9.2 Can the Model Replace Sensory Evaluation?

No. The test-set metrics show that the regression models are not reliable enough to replace human tasters:

- **R^2 is low:** only 40.3% (red) and 26.5% (white) of the variation in quality is explained, so 60–73% is unexplained.
- **The error is large relative to the scale:** the RMSE of 0.62 (red) and 0.75 (white) and the MAE of 0.50 and 0.59 mean that predictions are typically half a point to three-quarters of a point wrong, when more than 80–90% of wines lie within only three adjacent scores (5–7). A one-point error often crosses the boundary between an average and a good wine.
- **Extreme wines are missed:** the actual-versus-predicted plot shows that the best (8–9) and worst (3–4) wines are predicted as ordinary, which are exactly the cases a quality process should catch.

The models are better used as a low-cost **screening and decision-support tool**, for example to prioritise batches for tasting, while final quality decisions remain with sensory panels. The red wine model is the more dependable of the two.

9.3 Limitations

- **Limited predictive performance and linearity:** the models explain 40% and 27% of the variation, and linear regression assumes straight-line effects, so non-linear relationships and interactions are missed.
- **Subjective, imbalanced target:** quality scores come from human sensory assessment and cluster around 5–7, so the data contain little information about the best and worst wines.
- **Multicollinearity in white wine:** the strong correlation between residual sugar, density and alcohol makes the individual coefficients unreliable.
- **Duplicate records and a single data source:** 240 red and 937 white rows are duplicates. Because the train–test split was random, identical rows may appear in both

sets and slightly inflate the test performance. The results come from one region and have not been validated on independent data.

CHAPTER 10
CONCLUSION AND REFERENCES

10.1 Conclusion

This report presented two data-science case studies that follow the complete workflow of understanding data, exploring it, preprocessing, modelling, evaluating and interpreting the results.

In Part A, three clustering algorithms were applied to 3,999 EastWest Airlines customers. After standardisation, K-Means with K = 4 produced four balanced and meaningful segments: low-activity members (64%), credit-card earners (32%), frequent flyers (4%) and a very small elite group, with a silhouette score of 0.3092. Average and complete linkage obtained higher silhouette scores (0.7731 and 0.5986) but only by isolating a few extreme customers from one cluster of almost all members, and DBSCAN (eps = 2.5, minPts = 5) identified 98 outlying customers as noise. K-Means is therefore recommended for segmentation and DBSCAN for finding exceptional customers; the main lesson is that a high silhouette score is not enough without checking cluster sizes and business meaning.

In Part B, alcohol was the attribute most strongly and positively associated with wine quality, followed by volatile acidity (red) and density (white), both negative. Multiple linear regression reached R^2 = 0.4032 (red) and 0.2653 (white) on the test set, with RMSE of 0.62 and 0.75 quality points. Laboratory measurements of alcohol, volatile acidity and density can support preliminary screening, but the models are not accurate enough to replace sensory evaluation. Overall, the studies show that choosing the right evaluation measure and interpreting the model critically are as important as choosing the algorithm.

10.2 References

[1] EastWest Airlines customer loyalty dataset (EastWestAirlines.xlsx), provided with the course assignment on clustering analysis.

[2] P. Cortez, A. Cerdeira, F. Almeida, T. Matos and J. Reis, “Modeling wine preferences by data mining from physicochemical properties,” Decision Support Systems, vol. 47, no. 4, pp. 547–553, 2009.

[3] UCI Machine Learning Repository, “Wine Quality” dataset. https://archive.ics.uci.edu/dataset/186/wine+quality

[4] J. MacQueen, “Some methods for classification and analysis of multivariate observations,” Proc. 5th Berkeley Symposium on Mathematical Statistics and Probability, vol. 1, pp. 281–297, 1967.

[5] M. Ester, H.-P. Kriegel, J. Sander and X. Xu, “A density-based algorithm for discovering clusters in large spatial databases with noise,” Proc. KDD-96, pp. 226–231, 1996.

[6] P. J. Rousseeuw, “Silhouettes: a graphical aid to the interpretation and validation of cluster analysis,” Journal of Computational and Applied Mathematics, vol. 20, pp. 53–65, 1987.

[7] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.